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How Does Acknowledging Users' Preferences Impact AI's Ability to Make Conflicting Recommendations?

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ABSTRACT

Artificial intelligence (AI) decision support systems are crucial in modern decision-making processes. Their increasing human-like adaptability introduces challenges, especially when their recommendations, for whatever reason, need to conflict with user preferences. This study examines the communication strategies AI systems should employ when their recommendations conflict with user preferences. We explored this research question through a hypothetical future interface where ChatGPT offers travel recommendations populated on a map. An online survey-based experiment was conducted, presenting 160 participants with ChatGPT-generated travel recommendations displayed alongside Bing map visuals. We employed a mixed-method experimental design, combining both between-subjects and within-subjects approaches, to investigate the impact of conflicting recommendations and the acknowledgment of user preferences on the acceptance of these recommendations. This effect is especially pronounced when the AI system acknowledges users' preferences yet still offers conflicting recommendations to them. Contrary to the expectation that acknowledging users' preferences could buffer the impact of such conflicts, our observations indicate the contrary. The presence of conflict following acknowledgment of users' preferences, significantly causes a backfire effect, leading users to reject the recommendations. These findings underscore the need for consideration of recommendation delivery strategies in AI decision support systems and offer insights for designing future user interfaces and user experience research in the realm of recommendations provided by AI decision-support systems.

KEYWORDS

Acknowledgement; AI-assisted decision-making; recommender systems; explanations; artificial intelligence; Large Language Models (LLM); user experience; HCI

1. Introduction

AI-assisted decision making is increasingly common across diverse domains. The use of these systems can be found in diverse fields such as medicine (Hwang et al., 2019; Jussupow et al., 2021; Rajpurkar et al., 2022; Topol, 2019), education (Samarakou et al., 2018; Wambsganss et al., 2020), judiciary (Chilamkurthy et al., 2018; Sengupta & Dave, 2022), and financial assessment (Downen et al., 2024; Kou et al., 2019). Dabija and Vătămănescu (2023) presents thorough discussion on how AI has been incorporated into various domain. One major commonality is the use of natural language, where Becker et al. (2022) looks at generating natural language descriptions of nearly optimal strategies and showed that automatically generated decision aids significantly improved people's performance. A major contributor to this success is the effectiveness of deep learning algorithms. These systems, characterized by their expert-level accuracy, have proven to be effective in providing recommendations. Consequently, individuals increasingly rely on these AI-assisted systems to enhance their decision-making processes. However, even when AI-assisted systems have valuable correct advice for the decision being made, there may be barriers that prevent human users from accepting that advice. Indeed, studies (Komiak & Benbasat, 2006;

Lucas et al., 2018a, 2019) illustrated situations where users refuse to accept recommendations that are objectively correct or advisable. For instance: Lucas et al. (2018a) explored how cultural differences impact user acceptance of correct AI recommendations due to conflicts in values or personal beliefs. Therefore, the present study considers a new potential barrier, specifically: recommendations that differ from user preferences.

Several studies have sought to elucidate the psychological underpinnings of decision-making activities. Logg et al. (2019) revealed a tendency among humans to favor algorithmic over human advice, even when both are identical, thereby enhancing their reliance on technological systems. Moreover, the success of AI-assisted systems is contingent upon users' willingness to accept and act upon the system's recommendations, which, in turn, hinges on trust in the technology. Liu and Tao (2022) highlighted the pivotal role of trust in fostering user acceptance of these systems. This raises important considerations for the communication strategies AI systems should adopt when their recommendations conflict with user preferences. These strategies can be examined in the context of how generative artificial intelligence-based cognitive, emotional, and social processes can enhance decision-making and behavior (Peters et al., 2024).

While AI-assisted systems enhance prediction accuracy, the ultimate decision to embrace or disregard a recommended action rests with the human user. There have been several factors identified that can impede willingness to accept the recommendations from such systems. We define these as barriers—factors that result in the rejection of AI recommendations, even when there is an alignment between user preferences and the suggestions provided by AI. For example, users may hesitate or reject recommendations—even if they align with their preferences—due to systematic errors or biases stemming from data quality issues. Even when AI models offer precise recommendations, users show a tendency to trust human judgment over AI in decision-making scenarios where errors are noticeable. Conversely, algorithms are more likely to be trusted when their predictions are nearly perfect or when mistakes are not readily apparent, as noted by Dietvorst et al. (2015). Thus, errors serve as a significant barrier to the acceptance of algorithmic recommendations. Indeed, extensive research (Dzindolet et al., 2003; Lucas et al., 2017, 2018a, 2018b, 2019; Wang et al., 2013; Wiegmann et al., 2001) indicates that advice from agents prone to errors is less accepted than that from agents with high accuracy. Dzindolet et al. (2003) demonstrated that users distrusted reliable agents when they make errors, unless an explanation was provided regarding for potential errors. Furthermore, Wang et al. (2013) found users reacted with distrust towards agent that made conversational errors.

In addition to errors, an over-reliance on social norms for explanation or justification also hampers acceptance. Lucas et al. (2019) found that recommendations grounded in social norms, such as reciprocity, are less likely to be accepted compared to those justified with factual information. Their research highlights a general preference for advice supported by concrete information over social conventions.

Beyond errors and social norms, diminished forms of communication also present challenges, potentially leading to the rejection of recommendations that otherwise match individual preferences (Khashe et al., 2019, 2018, 2017). For example, compared to recommendations made via speech by an agent, the same recommendations made via text are less likely to be accepted. Likewise, compared to recommendations integrated as part of an agent's social dialogue such as "getting to know you" conversations, recommendations in the absence of social dialogue are less likely to be accepted.

Here, we propose a new barrier that could reduce users' willingness to accept recommendations. Specifically, we hypothesize that recommendations that conflict with the user preference can hinder users' tendency to accept the recommendations provided by AI-assisted systems. That is, the presence of a conflict between AI-generated recommendations and user preferences diminishes the likelihood of user acceptance. There can exist a tension between good advice and personalizing recommendations to match user preferences. This would mean that when users reject a recommendation because it conflicts with preferences, they may be making a mistake in not considering and accepting the personalized

advice. Here, mistake means users are not taking into account recommended personalized advice even if it aligns with their preferences. In spite of this potential tension between good advice and preferences, a lot of work has focused on personalizing support from AI systems (e.g., Komiak & Benbasat, 2006; Ma et al., 2017; Zhou & Han, 2019; Kim et al., 2021; Gao et al., 2023). For instance, Ma et al. (2017) and Zhou and Han (2019) proposed a user-preference-based collaborative filtering approach and a graph-based ranking method, respectively, to better incorporate user preferences in AI recommendations. Additionally, LLMs are being utilized to enhance user preference learning in Gao et al. (2023). Research by Komiak and Benbasat (2006) demonstrated that the perception of personalization significantly influences users' beliefs about the efficacy of recommendation agents. In the healthcare domain, Guo et al. (2016) discovered that personalization of mobile healthcare services significantly boosts user trust and acceptance.

While this research on personalization shows value in matching people's preferences, in the case that their preferences conflict with the good advice, rationally users should consider the better option even if it conflicts with their preference. Therefore, we examine the extent to which this conflict with their preference serves as a barrier to accepting the recommendation based on good advice.

Beyond attempting to establish that conflicting with users' preferences serves as a barrier to accepting a recommendation from an AI-assisted system, this work also explores potential opportunities to overcome these barriers. One effective technique is the acknowledgment of errors. For example, Tzeng (2004) and Kox et al. (2021) have shown that when AI agents admit their mistakes, the negative impact on user acceptance is mitigated. Moreover, incorporating apologetic text within the system can foster a more user-centered design, thereby enhancing users' positive perceptions of the system, especially in scenarios involving errors (Akgun et al., 2005; Park et al., 2012). In several studies, acknowledgment of errors, coupled with explanations, has been shown to be crucial for rebuilding trust in AI systems (Kox et al., 2021; Park et al., 2012). Research has also indicated that subjective ratings of agent performance improve when agents acknowledge their errors (Lucas et al., 2017, 2018b, 2019). Specifically, Lucas et al. (2017) discussed the significance of error acknowledgment by social robots, while Lucas et al. (2018b) found that effectively managing conversational errors through social dialogue can enhance user trust. Furthermore, Lucas et al. (2019) provided evidence of social tactics for acknowledging errors, highlighting that such interactions can significantly impact the acceptance of recommendations from AI agents. These interactions suggest that acknowledging errors can significantly impact the acceptance of recommendations provided by AI agents when they made errors. Like acknowledging errors, expressing regret can also serve as a way to overcome the barrier. For example, a study by Dzindolet et al. (2003) found that users tend to distrust even reliable advice from a system after observing errors; to rebuild trust, strategies that

provide explanations and express regret are effective, with the expression of regret being particularly crucial.

Additionally, studies (Lucas et al., 2017, 2018a, 2018b) consider “rapport building” as a strategy to overcome the barrier of errors. One might expect that having had a rapport-building or bonding conversation where the human and agent get to know one another might help the users “forgive” the AI when these errors start to occur. In fact, Lucas et al. (2018b) found this has the opposite effect: When AI engaged in the rapport-building conversation before doing a task with workers where it made errors, they were less likely to accept the advice of the agent than when they did not have this conversation first. This effect of backfiring in rapport-building occurred even though the AI malfunctioned in superficial ways (e.g., repeating answers or requests), despite its recommendations to the user being 100% correct. The reason behind this backfire effect could be reactance, or psychological “pushback” when our expectations are violated. De Visser et al. (2016) and de Visser et al. (2012) explain that higher expectations make people less forgiving of machines’ errors than those made by humans. The rapport-building conversation went quite smoothly in Lucas et al. (2018b), which raised expectations for users. Therefore, when participants experienced this effortless conversation followed by errors during the task, it effectively dashed their hopes of good performance—even though the agent was correct in its recommendations during the task. Given the strong contrast between the initial rapport-building conversation and the subsequent errors, users likely experienced reactance when they built rapport with the AI first compared to those that did not (and thus their expectations were not built up so much). Indeed, additional experimental conditions supported this interpretation (Lucas et al., 2018b). Users who experienced the rapport-building conversation later, after that the task was filled with superficial errors, accepted the recommendation of the AI to the same extent as those who did not build rapport. Because the rapport-building conversation occurred after the task, it did not set them up for a contrast effect (like those who had the conversation first), and therefore user acceptance was not negatively affected. If anything, having a rapport-building conversation after helped to restore some of the trust that was lost due to the errors.

While rapport building is an equivocal approach to addressing barriers such as errors, the literature is more consistent with regard to acknowledging errors. Accordingly, we adopt this approach in the current study and examine whether there is a benefit of acknowledging when the recommendations conflict with user preferences. That is, we test the extent to which such acknowledgment can restore willingness to accept the recommendations when they conflict with preferences.

We consider acknowledgment because it has established efficacy in interpersonal communication. Acknowledgment not only facilitates mutual understanding between humans and AI systems but also improves human perceptions of interaction quality in Human-AI interactions (Paranjape & Manning, 2021). It serves as a mechanism to ensure that

information transmission includes an understanding of the conveyed content, thereby synchronizing interpretation and enhancing the alignment between communicators.

Despite the extensive research on error acknowledgment, research has not specifically addressed acknowledging when conflict occurs between recommendation and user preferences. We explore the use of acknowledgment as a strategy to overcome such barriers. By drawing parallels with the benefits observed in error acknowledgment scenarios, we hypothesize that acknowledgment may mitigate conflicts and increase the acceptance of AI agents.

2. Methodology

There are two primary aims of this study. First, we evaluate a novel barrier preventing the acceptance of AI recommendations. Namely, the conflict between the AI’s recommendations and the user’s preferences. Second, we explore whether this barrier can be circumvented by acknowledging user’s preferences. To help address these aims, we define the following variables:

Conflict	Conflict refers to a recommendations where there is a mismatch between what the user prefers and what the AI agent recommends.
Acknowledgement	We define acknowledgement as explicitly recognizing a user’s preference. This is a natural and effective way to convey understanding and to frame alternative recommendations (Clark & Brennan, 1991). Acknowledgement of a user-specified preference can either exist or not exist (see Table 1 for an example).
Conflict Position	This variable indicates where a conflicting recommendation took place. This helps us understand whether the degree of preference affects acceptance of misaligned recommendations. The recommendation can either (1) conflict with the subject’s stronger preference (2) or conflict with the subject’s weaker preference.

2.1. Participant recruitment

The experiment was conducted with 160 participants, between 18 and 45 years from UK and USA ($M = 31.84$ years, $SD = 6.86$) from Prolific. Participants involved in the survey were a balanced sample: equal percentages of male (79 male participants) and female (81 female participants) with an approval rate exceeding 98%, where approval rate signifies those participants who completed their tasks accurately and honestly based on the research instructions in the prolific. Informed consent was obtained from all participants and each participant was paid 12\$ per hour. All personnel conducting the experiment have the necessary IRB training certification and study approvals.

2.2. Problem setting

We utilized ChatGPT (Brown et al., 2020) to generate the recommendations for each condition. Each response is based on an entirely new session with the first interaction being a prompt of interest. These generated responses are then

Table 1. Example prompt style and synthesized response.

Prompt style	Example
Give reasons to go to location A. (<i>Non-acknowledging, non-conflicting condition.</i>)	House Without A Key is an iconic restaurant and lounge in the Halekulani Hotel in Waikiki, Hawaii. It's best known for its poolside location with beachfront views and sunset vistas. Your cares will melt away when you are immersed in the ambiance! Here are three reasons to choose House Without A Key: (1) Unique Atmosphere: House Without A Key offers a <i>unique and inviting ambiance</i> that combines the charm of a picturesque setting with good food. (2) Cultural Experience: <i>House Without A Key is not just a place to dine; it's an experience.</i> With live Hawaiian music and hula performances, visitors get a taste of Honolulu's rich cultural heritage. You can focus on the experience, which can be a refreshing change of pace. (3) After the Meal: First, have a meal at House Without A Key, which provides a comfortable and enjoyable dining experience. After dining at House Without A Key, one can choose to head into the heart of Waikiki. Visitors can explore the myriad of shops, bars, clubs, and other attractions that Waikiki has to offer. In essence, House Without A Key provides a unique blend of beachfront beauty, cultural richness, and proximity to attractions, making it a worthwhile recommendation.
Suggest someone to visit Moanalua Valley Trail when they prefer to go to the beach. (<i>Acknowledging and conflicting condition.</i>)	<i>While the beach offers its own unique charm</i> , if you're open to a slightly different experience away from the sandy shores, Moanalua Valley Trail might be the perfect alternative for you. The trail presents a fantastic opportunity to infuse your beach day with a touch of adventure and nature exploration. Here are three reasons why: (1) Change of Scenery: <i>While the rhythmic sound of waves and sunbathing has its appeal</i> , the lush greenery, gentle streams, and expansive valley views of Moanalua present a refreshing change. (2) Cooling Off: <i>Hiking in the valley can often be cooler than lying on a sunny beach.</i> The canopy of trees provides shade, making the experience more comfortable, especially if you're looking to escape the intense midday sun. (3) Accessible Adventure: <i>While the beach offers relaxation</i> , Moanalua Valley Trail gives you a chance to experience the rewards of being a bit more active, offering a very fulfilling, but reasonable challenge. So, for a change of pace that complements your love for the beach, consider giving the Moanalua Valley Trail a chance—a day that transitions from sandy shores to the lush heart of the island.

LLMs tend to point out the main *features* of the place regardless of the different styles and tones used in prompting. We began with general prompts like “Give reasons” or “Suggest” and then moved towards contrasting prompts. In the contrasting prompts, preference contrast with its feature, like “Moanalua Valley Trail” is known for hiking. The feature of place contrast with the user preference i.e., beach. Here, LLM acknowledges user's preference and list out few points regarding the features of the place presented in the prompt, which is similar to general prompts like “Give reasons” or “Suggest” for simplicity, we shortlisted three rationales to analyze. Furthermore, in our research, we tend to use the general prompts provided with a fixed number of reasons.

normalized so that they use the common language across all conditions for experimental control and validity. Normalization measures include things such as assigning the number of reasons, using the most common phrasing across synthesized content, standardizing layout, etc.

Initially, we investigated how different verbs (give, suggest, ask, tell, etc.) affect the output response. For every verb used, the generated response tend to provide the characteristics of the location in the prompt. For example: [*Suggest, ask, tell, advise*] someone to visit Pu'uhonua O Honaunau historical park. In various verb choices, the responses consistently accentuate the characteristics of the specified locations in the prompt. We further experimented by adding user preferences in the prompt. For example, *Suggest someone to go to A if they like [some characteristic of A]*. We found that in these instances, LLMs tend to emphasize these properties in the provided response (Perez et al., 2023). We can create conflicting scenarios by telling the LLM the preference given by the subject using the form *Suggest someone to go to [A] when they prefer to go to [some characteristic of B]*. For example: *Suggest someone to visit Moanalua Valley Trail when they prefer to go to the beach*. Please see Table 1 for examples of prompts and responses.

Through this process, we found that when LLMs are asked to generate a response conflicting with user preference, the default approach is to first acknowledge a user's preference. From a mathematical optimization perspective, LLMs are trained to follow most-likely next token statistics. Thus, it provides an indication that acknowledgement is likely a common method to employ when proposing conflicting suggestions in online or text-based interactions.

Using this knowledge, we standardized each LLM response. We did not use a unique LLM response for each experimental condition as there are slight differences with each response synthesized. Furthermore, we wanted to develop a controlled experiment to examine the variables of acknowledgment, conflict, and user preference regardless of what underlying LLM model was being used. Table 1 provides prompt styles and examples that were used as recommendation.

2.3. Experimental design

This study aims to explore the role of conflict between AI-generated recommendations and user preferences as a significant barrier. Moreover, we examine the potential of leveraging acknowledgement to mitigate this conflict barrier. To assess whether acknowledging the conflict could improve user satisfaction, we integrated this acknowledgement directly into the recommendations. Specifically, the inclusion of the acknowledgement factor was facilitated through the generation of responses using Large Language Models (LLMs) in a controlled manner. The methodology employed for generating recommendations via LLMs is elaborated upon in Section 2.2.

We designed a trip planning scenario to explore the extent to which user's accept the recommendations when the AI recommendation acknowledges user's preference and when it conflicts with the user preference. We chose trip planning because it provides real-life context to participants (Kurashima et al., 2010; Lu et al., 2012). Furthermore, it provides a straightforward way for evaluation. For our

Table 2. Experimental design conditions.

Strong preferences (Honolulu)		
Category		
Restaurant	A: Pool Side	B: City View
Activity	C: Beach	D: Hiking
Weak preferences (Honolulu)		
Category		
Restaurant	E: Volcano View	F: Ocean View
Activity	G: Cultural/historical Tour	H: Recreational Activities

Each participants states their preference between above two categories. Conflict position is within-subjects variable, that indicated where conflict takes place based on the user preference. Acknowledgement and conflict are between-subjects conditions.

destination, we chose Hawaii as it is a well-known place with a diverse set of vacation activities.

2.3.1. Experimental conditions

We designed a $2 \times 2 \times 2$ mixed factorial experiment. The between-subjects variables are conflict (conflict vs. no conflict) and acknowledgment (acknowledging user preference vs. not acknowledging user preference). The between-subjects condition requires the users to specify a preference for a particular vacation category they would like to participate in. This means that there are two positions where conflict can take place. We define a third conflict position within-subjects variable to understand if where the conflict occurs has an effect. This within-subjects variable can conflict with the strong preference or with the weak preference. All participants received conflict positions in a set order: they first received the conflict with strong preference where recommendations were made for activities and restaurants in Honolulu, then they received the conflict with weak preference where recommendations were made for activities and restaurants on the Big Island. So, there are two locations in Hawaii for within-subjects variable: Honolulu and Big Island.

2.3.2. Experimental procedure

We first asked participants what their preferences were between two vacation categories: (1) types of activities and (2) types of restaurants (see Table 2). Participants first indicated whether they preferred restaurants at the pool or with a view of the city. They were then asked if they preferred to go to the beach or on a hiking expedition. Importantly, we asked participants which of these preferences were stronger.

Once the participants indicated their preferences, they received a random recommendation based on this preference. The first condition that the participant receives is always the stronger of the previously indicated choices. From here, the recommendation may or may not align with the participants preference depending on whether it was a conflicting condition or not. In the case of a conflicting condition, the conflict takes place with the participants' strong preference. Specifically, in conflict with strong preferences, if participants stated that the activity decision was more important than the restaurant decision, then the AI-recommendation system would give a recommendation that conflicts with their preferences of the activity. That is, if they said they preferred beach over hiking, it would make recommendations for a place to hike. If instead, they preferred hiking over beach, it would

make recommendations for a beach. On the other hand, whatever the participant's weaker preference was (activity or restaurant), the AI-recommendation system made a recommendation that was consistent with their preference. For example, if their weaker preference was for the restaurant and they preferred a city view, it would recommend a restaurant with a city view. This represents the workflow for the conflict with strong preference experiment condition.

In the second phase of the experiment, or the conflict with weak preference condition, participants first indicated whether they preferred restaurants at the ocean or with a view of a volcano. We then ask them, for an activity, whether they prefer going to the cultural/historical tour or a recreational activity. Again, we then asked participants which of these preferences were stronger. In this case, the conflict takes place with the participants' weaker preference. For example, if participants stated that the activity category was less important than the restaurant decision, then the AI-recommendation system would give a recommendation that conflicts with their preferences of the activity. That is, if they said they preferred cultural/historical tour over recreational activity, it would make recommendations for a recreational activity. If instead, they referred recreational activity over cultural/historical tour, it would make recommendations for a cultural/historical tour. A summary of the possible experiment conditions can be inferred from Table 2.

2.3.3. Between-subjects variables: Acknowledgement and conflict

The acknowledgement variable was also manipulated. Because both conflict and acknowledgement variables were between subject variables, this resulted in four experimental conditions:

1. The combination of acknowledging and conflicting preference: Participants' stated preferences are *acknowledged* however they received recommendations that did not align with their preferences. AI recommender system tries to explain how the alternative option (e.g., Moanalua Valley hiking trail) is actually really good for people who like the beach (see Table 1).
2. The combination of non-acknowledging and conflicting preference: Participants' stated preferences are *not explicitly acknowledged* and they received recommendations that were *different* than stated preferences. AI recommender system tries to explain how the alternative option (e.g., Moanalua Valley hiking trail) is really good presumably written for an audience who like hiking (see Table 1).
3. The combination of acknowledging and non-conflicting preference: Participants under acknowledgement in conflict condition received recommendations that aligned with their preferences and their preferences was acknowledged. AI recommender system explains how the option (e.g., Moanalua Valley hiking trail) is actually great for people who like hiking, acknowledging that it knows the participant likes hiking.

Table 3. Experimental design flow table.

Conflict position	Description	Location	Conflict condition
Conflict with strong preference	Initially, participants are presented with strong preference conflict position.	Honolulu	Conflict occurs only for the decision where the participant has a strong preference. If activity is chosen by the participant, then there is conflict in strong preference i.e. activity. No conflict for the decision with a weak preference.
Conflict with weak preference	Once the 'conflict with strong preference' condition is complete, participants are presented with the condition where there is a conflict with the weak preference	Big Island	Conflict occurs only for the decision where the participant has a weak preference. If activity is chosen by the participant, then there is conflict in weak preference i.e. restaurant. No conflict for the decision with a strong preference.

This table provides summarized view of conflict position that was conducted in the experiment.

- The combination of non-acknowledging and non-conflicting preference: Participants under disregarding acknowledgement in conflict condition received recommendations that aligned with their preferences and their preferences was not acknowledged. AI recommender system explains how the option (e.g., Moanalua Valley hiking trail) is actually great, with the emphasis on preferring hiking removed.

An example of this is provided in Table 1. The four conditions above are present in both locations resulting in eight total conditions.

2.3.4. Data collected

Throughout the experiment, in both within-subjects condition, participants rated their preference strength for each category. Within each category, two options were presented, as described in Table 2. Participants indicated their preference among these options using a 7-point Likert scale. In this scale, a rating of 7 represents that their chosen option is the one they preferred most, while a rating of 1 indicates that both options are perceived as equal. Following this, participants were asked to rate their preference between the two categories using a binary Likert scale, indicating which category they preferred more. The category they preferred more is termed as strong preference and the other one as weak preference. Table 3 summarizes the details of the conflicting recommendation for the user selected preference in the two within-subjects condition.

3. Results

We present results analyzing the effect and perceptions of a system that conflicts with participants' preferences. First, we discuss how results are analysed (3.1). Second, we present our results examining differences based on conditions to see when users are willing to accept (or reject) recommended activities (3.2). Third, we examine general perceptions of the system to provide insight for future design (3.3).

3.1. Data recoding

As described above in Section 2.3.2, in both within-subjects condition, people decide on both the activity and the

restaurant. While these are two dependent variables, activity and restaurant, that occur for each conflict position, we do *not* analyze it by activity and restaurant. Rather, for interpretability, we recode the data in order to be able to ignore the decision content (activity vs. restaurant) and focus the analysis on what matters to our research question: the decision for which participants had a strong preference in a given conflict position (whether that's activity decision or restaurant decision) vs. the decision for which participants had a weak preference (the other decision remaining).

In conflict with strong preference—Honolulu, even in the *conflict condition*, there is *no conflict between the recommendation* and the participant's preference for the decisions for which *participants had a weaker preference*. The conflicting condition only differs (i.e., presents a recommendation that conflicts with the participant's preference) for the decisions for which participants had a stronger opinion in conflict with strong preference.

Likewise, in conflict with weak preference—Big Island, even in the *conflict condition*, there is *no conflict between the recommendation* and the participant's preference for the decisions for which *participants had a stronger preference*. The conflicting condition only differs (i.e., presents a recommendation that conflicts with the participant's preference) for the decisions for which participants had a weaker preference in conflict with weak preference.

3.2. Acceptance of recommendation

First, participants decided to accept the recommendation made by the AI or not. In the first within-subjects condition, (conflict with strong preferences)—we examined the decision for which participants had a strong preference (activity decision if they stated they had a stronger opinion for activity or restaurant decision when they stated they had a stronger opinion for the restaurant). To test the effect of conflict and acknowledgement, log linear analysis was used. We found that the conflict has only an effect on decisions. Participants are less likely to accept (63 accept vs. 19 reject) in the conflicting condition than the non-conflicting condition (72 accept vs. 6 reject; $G^2(1, N=160) = 7.62, p=0.006$). There was no other significant effects $G^2 < 7.7, ps > 0.10$). Still considering the strong preference condition,

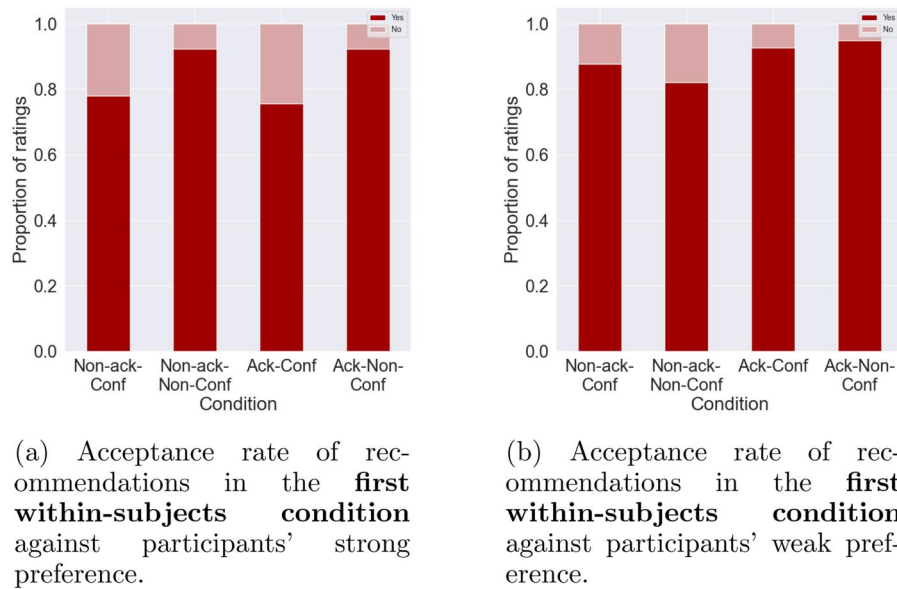


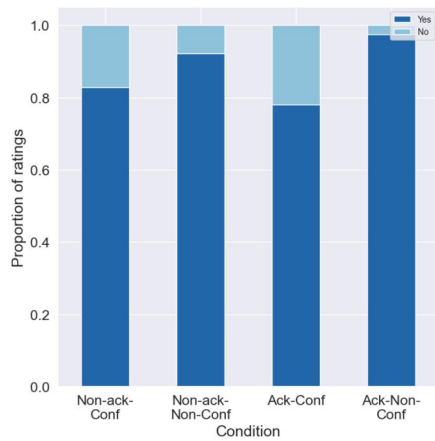
Figure 1. The provided bar chart illustrates user acceptance of recommendation in the *first within-subjects condition*. The labels on the y-axis represents between-subjects variable which are succinctly abbreviated as: *acknowledging and conflicting preference* (Ack-Conf), *non-acknowledging and conflicting preference* (Non-Ack-Conf), *acknowledging and non-conflicting preference* (Ack-Non-Conf), *non-acknowledging and non-conflicting preference* (Non-Ack-Non-Conf).

we next look at the decisions for which participants had a weak preference (activity decision if they stated they had a stronger preference for restaurant or restaurant decision when they stated they had a stronger preference for the activity). There were no significant effects (G^2 s < 3.3 , p s > 0.07). Participants generally accepted such recommendations (143 accept vs. 17 reject) as there was no conflict even in the conflicting condition because for this within-subjects condition (Honolulu), the conflict occurs only with the strong preference, not the weak one. We show a plot of this in Figure 1.

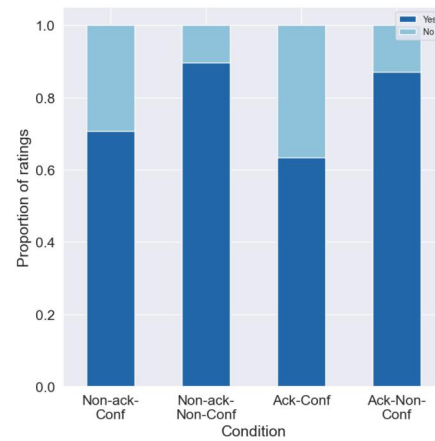
In the second within-subjects condition, (the second conflict position—conflict with weak preferences)—we examined the decision for which participants had a weak preference shown in Figure 2. Examining the effect of conflict and acknowledgment, there was only an effect of conflict, indicating that participants were less likely to accept the recommendation in the conflicting condition, than the non-conflicting condition (69 accept vs. 9 reject; G^2 (1, $N = 160$) = 10.92, $p = 0.001$). However, a significant interaction occurred (G^2 (1, $N = 160$) = 11.56, $p = 0.02$), revealing that acknowledging user preferences backfired when there is a conflict. No differences were observed in the acknowledging preferences, where participants generally accept the recommendation (34 accept vs. 5 reject in conflict, and 35 accept vs. 4 reject in non-conflicting). However, in the conflicting condition, users were less likely to accept recommendations when the AI acknowledged their preferences (26 accept vs. 15 reject), than in the non-acknowledging condition (29 accept vs. 12 reject). When users preferences were acknowledged in the conflicting condition, the AI tried to make an option against their preferences sound like it actually meets their preferences. Whereas, in the non-acknowledging condition, the AI just explains why the option would be good for someone who prefers it. No other effects reached statistical significance (G^2 s < 0.6 , p s > 0.43).

In the second within-subjects condition, (the second conflict position—conflict with weak preferences)—we examined the decision for which participants had a strong preference (activity decision if they stated they had a stronger preference for activity or restaurant decision when they stated they had a stronger preference for the restaurant). Examining the effect of conflict and acknowledgment, there was an effect of conflict (66 accept vs. 16 reject in conflicting, and 74 accept vs. 4 reject in non-conflicting; G^2 (1, $N = 160$) = 8.08, $p = 0.005$). Even though there was no conflict for this particular recommendation (i.e., it matched their preference because the choice for the strong preference in this within-subjects condition when conflict was for the weak preference, not the strong one), participants were still less likely to accept when they were assigned to the conflicting condition and therefore, two of their other recommendations were previously conflicting, compared to those assigned to the non-conflicting condition and therefore had not experienced any conflict with the preferences. A marginally significant interaction (G^2 (1, $N = 160$) = 9.52, $p = 0.05$) suggested that the backfire effect for acknowledging user preferences when the recommendation conflicts can bleed into decisions where there is no longer any conflict. When there had never been any conflict (non-conflicting condition), acknowledging user preferences had no effect (38 accept vs. 1 reject in the acknowledging condition and 36 accept vs. 3 reject in the non-acknowledging condition).

Even though there was no conflict in this particular recommendation, users in the conflicting condition (who had experienced conflict previously) were less likely to accept when the AI acknowledged their preferences (32 accept vs. 9 reject) compared to when preferences were not acknowledged (34 accept vs. 7 reject). No other effects reached statistical significance (G^2 s < 0.2 , p s > 0.88). For the second conflict position, conflict with weak preference—Big Island, even in the conflict condition, there is no conflict between



(a) Acceptance rate of recommendations in the **second within-subjects condition** against participants' strong preference.



(b) Acceptance rate of recommendations in the **second within-subjects condition** against participants' weak preference.

Figure 2. The provided bar chart illustrates user acceptance of recommendation in the *second within-subjects condition*. The labels on the y-axis represents between-subjects variable which are succinctly abbreviated as: *acknowledging and conflicting preference* (Ack-Conf) *non-acknowledging and conflicting preference* (Non-Ack-Conf), *acknowledging and non-conflicting preference* (Ack-Non-Conf), *non-acknowledging and non-conflicting preference* (Non-Ack-Non-Conf).

the recommendation and the participant's preference for the decisions for which participants had a stronger preference. Accordingly, there is no conflict between the recommendation and the participant's preference for this choice in either the conflict or in the no conflict conditions; the conflict conditions thus do not differ from each other for the decisions for which participants had a stronger opinion in this conflict position, the conflicting condition only differs (i.e., presents a recommendation that conflicts with the participant's preference) for the decisions for which participants had a weaker opinion in this conflict position. In Figure 2 provides the acceptance rate of recommendation with four different conditions.

3.3. Perceptions

After making their decisions and ratings of the recommendations, participants reported their overall perceptions of the AI in case of conflict barrier. We adopted DeVault et al. (2014) scale to examine the perceived positivity of the user experience. There is a significant interaction between the two factors ($F(1,156)=4.13$, $p=0.04$; that qualified a main effect of conflicting condition ($F(1,156) = 5.16$, $p=0.03$), but main effect for acknowledging condition did not approach significance ($F(1,156) = 1.70$, $p=0.19$). As expected, when the recommendation aligns with the user's preferences, acknowledging those preferences by customizing the recommendation to highlight how it matches their preference resulted in the system being rated more highly ($M=4.16$, $SE=0.10$) than when it gave a generic recommendation that did not emphasize how it matched their preferences ($M=3.83$, $SE=0.10$).

In contrast, acknowledging user preferences when there is a conflict, and attempting to make an option contrary to their preferences sound like it aligns with their preferences,

did not enhance the user experience. When the recommendation conflicted with the user's preferences, they rated their experience as, if anything, less positive when it acknowledged their preferences ($M=3.73$, $SE=0.10$) than if when it did not acknowledge their preferences and just explained why the option would be good for someone who preferred that kind of location ($M=3.81$, $SE=0.10$).

To better understand the perceptions underlying the ineffectiveness of acknowledging user preferences in the case when the recommendation conflicts with them, we adopted the approach from (Gilani et al., 2016). Specifically, we used the person perception items to see how the AI was perceived on each of the 32 dimensions individually; here we report all significant interactions (main effects are not reported as they cannot help us to understand the above interaction effects). First, when there was no conflict between the recommendation and user's preferences, the AI was seen as being more "permissive" (or less stern) when it acknowledged those preferences ($M=4.15$, $SE=0.19$) than when it did not ($M=4.74$, $SE=0.19$); in contrast, when the recommendation conflicted with the user's preferences, the AI was seen as less permissive (or more stern) when it acknowledged those preferences ($M=3.98$, $SE=0.18$) than when it just focused on the benefits of the conflicting location without highlighting that it conflicted with their preferences ($M=4.42$, $SE=0.18$; $F(1,155) = 7.87$, $p=0.006$). Along these same lines, when there was no conflict with user's preferences, the AI was seen as more relaxed (less tense) when it acknowledged those preferences ($M=4.30$, $SE=0.20$) than when it did not ($M=4.03$, $SE=0.20$); but there was no benefit of acknowledging preferences when the AI's recommendation conflicted with the user's preferences ($M=4.42$, $SE=0.19$ vs. $M=4.44$, $SE=0.19$; $F(1,156) = 3.79$, $p=0.05$). However, all effects failed to approach significance for the negative person perception items

($F_s < 3.14$, $p_s > 0.08$), as well as the scales for helpfulness and trustworthiness ($F_s < 3.00$, $p_s > 0.09$).

4. Discussion and limitations

Designing effective explanations for collaborative Human-AI decision-making becomes even more critical as AI reasoning capabilities improve. Furthermore, we need to understand how humans can be influenced by the social factors of AI recommendations to encourage effective decision-making and to reduce the impact of human behavioral biases. This is important because in many situations, for example when making optimal medical decisions, an AI may know more than a human and disagree with their preference. For instance, Arunthavanathan et al. (2024) examined this conflict particularly in process system operations and the risks posed by it. In our work, we introduced the presence of acknowledgement to overcome this conflict barrier; that is, we tested whether acknowledging the conflict could reduce the negative impact on taking an AI's recommendations. The initial hypothesis posited that users would hold a more favorable perception of the system when their preferences were acknowledged with detailed descriptions. However, contrary to these expectations, there was no discernible benefit in acknowledging preferences to overcome the conflict barrier. In fact, if anything, there was a backfire effect for acknowledging preferences when AI's recommendations conflicted with the user preference. This unexpected outcome can be attributed to psychological reactance (Brehm, 1966), which suggests that individuals, when sensing a threat to their autonomy or control by the advice, are inclined to push back. In essence, the negative impact of acknowledgment may stem from users' psychological reactance triggered by perceived threats to their autonomy. In a study (Bensley & Wu, 1991), it was found that strongly worded messages can induce reactance. Drawing parallels to these findings, acknowledging conflicting recommendations could have induced reactance because the messages were more strongly worded. Overall our results suggest that it is necessary to carefully frame recommendations when the AI's recommendation contradicts with the user preference.

On the other hand, it could be that acknowledgement was unable to overcome the conflict barrier either due to lack of perceived authenticity or due to behavioral biases such as the confirmation bias. Beyond reactance, another explanation for our findings might be that the acknowledgment might not have been perceived as genuine. If that was the case, users could feel that their preferences were overlooked or undervalued. This absence of genuine acknowledgment would then explain the hesitancy to accept the recommendation when it was conflicting. Finally, confirmation bias might also have played a role (Nickerson, 1998). This suggest that future work can try to manipulate how genuine an acknowledgment sounds to see if this is an issue with current text generated by GPT. Participants, influenced by confirmation bias, tend to favor information aligning with their existing beliefs or preferences, such as the specific kind of place they want to go on a trip. This

inclination might be particularly triggered by acknowledging the preferences because it reminds them of their initial preferences. More generally, we leave this work as a building block for future work in examining some of these other factors to overcome conflicting decision barriers.

While our results underscore the importance of carefully framing recommendations that contradict user preference, there are some methodological limitations. This study is limited by the scope of the experimental stimuli. A key property of this is how consequential the decision making is—which is low for vacation trip planning (compared to medical decision-making, for example). Extending this to additional tasks of varying consequence may yield different results. Also, a possible confound is that our first and second preference (stronger vs. weaker) stimuli were not randomized due to the complexity of the experimental design; and that a more complex design with counterbalancing would have required a larger sample size, which was not pragmatically feasible at this time due to limited funding. We interpret our findings as due to the effect of order, as that made sense given the pattern of results. However, it is possible that some aspect of -specifically- a weaker preference following a stronger preference may have instead driven the effects. However, this result is interesting in that it suggests that perceptions of AIs are sticky and warrants future investigation. Lastly, we are assuming that the recommendation could in fact be a better or a valid alternative answer to a user's preference. In previous studies, such as ones that use the NASA lunar survival task (NASA, 1999), AI's recommendations reflect the correct or best answer. However, in our trip planning task, we have not established unequivocally that this is in fact the case. Future work should explore how this fundamental difference can impact decision making processes. Hence, we advocate for continued exploration and research in this dynamic area, aiming to enhance the user experience and harness the full potential of AI-integrated decision support systems.

Given these limitations, particularly the potential influence of order effects and defined recommendations, our findings suggest that how AI recommendations are framed is crucial. Additionally, Chong et al. (2022), found that self human confidence had significantly impacts the adoption of AI recommendation. In closely related research, Ebermann et al. (2023) showed that providing explanation improved user acceptance in conflict scenarios. Similarly, studies on empathy and interaction, such as those by Pelau et al. (2021) and Fu et al. (2022), indicate that these factors can enhance acceptance of AI agents. Future work should explore how incorporating empathy and detailed explanations in conflicting recommendations might improve user acceptance.

Despite these limitations, our findings still offer valuable insights into the design of AI agents, particularly in how they handle decision-making interactions with humans. Our results suggest that acknowledging the user preferences in overcoming conflict backfired, indicating that agents should be carefully designed by considering how recommendations are framed. AI agents are being adopted into different

sectors to assist humans in accounting their preferences. For example: Eliasson et al. (2017) and Shirk et al. (2017) have integrated patient preference in medical decision making for treatment purposes; and Scherr et al. (2017) reported that expert treatment decision was accepted despite the patients' preferences. Zheng (2019) has proposed the solution as preference correction in educational recommendations where instructors and student have differing preferences. Similarly, AI agents might not always align with the user preferences. It is necessary to address AI agents' recommendations that could be accepted by users despite being conflicted with their preferences. We explored in acknowledging the user preferences to overcome the barrier and found this strategy backfired. Therefore, we urge researchers to carefully frame the recommendations and explore other strategies like detailed explanation regarding the conflicting recommendation in overcoming the conflict. Further work should explore strategies to mitigate conflicts in high-stakes environments like healthcare or autonomous driving, ensuring that AI agent recommendations are accepted by users even when they conflict with their preferences.

5. Conclusion

The prevalent use of AI across various domains has highlighted the need to understand the challenges associated with recommendations, particularly when these recommendations conflict with user preferences. This study investigated how conflict with user preferences can impact acceptance of an AI agent's recommendation in the context of a trip planning scenario. Conflict with preferences was found, as expected, to be a barrier to acceptance of AI's recommendations. We also introduced a possible remedy to this barrier: acknowledging that the preference conflicts with the recommendation, while explaining how the plan for the trip could still meet the user's needs despite this conflict. Our results revealed that, if anything, this strategy backfired, further reducing acceptance of the AI's recommendation. The results underscore the importance of carefully framing recommendations that contradict user preference, as well as exploring alternative strategies to mitigate resistance to conflicting advice. Future research should focus on developing and testing new methods to enhance the acceptance of AI recommendations, particularly in scenarios where preference conflicts are unavoidable.

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